GELOS
ENTERPRISES



Website Development Policy and Procedure

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1. About this policy and procedure

1.1 Document purpose

This document describes organisational policies and procedures applicable in developing dynamic websites. It includes organisational requirements, cyber security protocols and best practice industry standards.

1.2 Scope

This policy and procedure applies to all the members of the website development team, including designers, analysts, developers and Team Leader.

2. Best practice guidelines

Best practice in website design and development includes applying standards for building and rendering web pages.

2.1 W3C standards

The World Wide Web Consortium (W3C) defines the standards for website development, which all websites developed by Gelos should follow.

2.1.1 Use of web standards

W3C promotes the use of standardised technologies to ensure consistency and interoperability across the web. These include:

- HTML for structure
- CSS for presentation
- JavaScript for interactivity
- SVG, XML, WebRTC and various APIs for extended functionality.

These standards ensure that websites work reliably across different browsers and devices.

2.1.2 Accessibility

Web content must be accessible to all users, including those with disabilities. W3C's Web Content Accessibility Guidelines (WCAG) provide detailed criteria to:

- ensure text alternatives for non-text content
- make content navigable via keyboard
- provide sufficient contrast and readable fonts
- support screen readers and assistive technologies.

2.1.3 Internationalisation

Websites should support multiple languages and cultural formats. W3C encourages:

- Unicode encoding (UTF-8)
- locale-aware formatting (dates, numbers and so on)
- right-to-left and left-to-right text support.

2.1.4 Privacy and security

W3C standards emphasise secure and private user experiences. Best practices include:

- HTTPS for secure communication
- secure handling of user data
- use of authentication and encryption protocols.

2.1.5 Device independence

Websites should be usable on any device - desktop, tablet, mobile, or assistive technology. This includes:

- responsive design using CSS media queries
- flexible layouts and scalable assets
- avoiding fixed dimensions or device-specific code.

2.1.6 Interoperability

W3C standards ensure that web technologies work together seamlessly. Developers should:

- use valid, semantic HTML
- avoid proprietary or deprecated features
- test across multiple browsers and platforms.

2.2 CSS design principles

Gelos websites must follow CSS design principles to ensure efficiency, consistency and ease of use.

2.2.1 Consistency

Maintain a consistent visual style across all elements and pages of a website or application. This includes consistent typography, colour palettes, spacing and component styling.

2.2.2 Maintainability

Write CSS that is easy to understand, modify and extend. This involves using clear naming conventions, modularisation and avoiding being too specific.

2.2.3 Responsiveness

Design CSS to adapt seamlessly to various screen sizes and devices, ensuring optimal viewing and interaction regardless of the user's device. This is typically achieved through media queries, flexible layouts (Flexbox, Grid) and fluid units.

2.2.4 Performance

Optimise CSS for fast loading times and smooth rendering. This includes minimising file size, avoiding unnecessary complexity and leveraging browser rendering optimisations.

2.2.5 Accessibility

Ensure that the design is usable by individuals with disabilities. This involves considerations for colour contrast, keyboard navigation, focus indicators and semantic CSS.

2.2.6 Separation of concerns

Keep CSS focused solely on presentation, separating it from HTML (structure) and JavaScript (behaviour). This promotes cleaner code and easier maintenance.

2.2.7 Reusability

Create reusable CSS components and patterns to reduce code duplication and enhance consistency. This can be achieved through techniques like component-based styling, utility classes and using external style sheets.

2.2.8 Simplicity

Strive for the simplest possible solution to achieve the desired visual outcome, avoiding overly complex or convoluted CSS. Do not add unnecessary elements or attributes; use existing elements as selectors, wherever possible.

3. Procedures

The following organisational procedures are specified to provide clear criteria for decision making processes and how website design and development work should be carried out.

3.1 Design and diagramming

Design and diagramming tasks should use the following criteria:

- Diagramming modelling should use UML 2.5, where applicable.
- User interface (UI) storyboarding should use low-fidelity wireframes.

3.2 Authentication and security

Users should be authenticated and authorised to access systems. That is, access to systems should be controlled by using different access levels for different user types.

For information on password requirements refer to the Password policy in the Gelos User Account Policy (pdf).

3.3 User interface requirements

All developed or adopted user interfaces should be intuitive, provide efficient access for usage (including touch-screen devices) and meet expected organisational requirements for accessibility.

Menus and navigation should be appropriately presented using best practice guidelines and styles.

3.4 User data and privacy

All user data should be stored and accessed in accordance with the Australian Privacy Principles.

User data is only to be displayed to authorised and relevant users.

Passwords are to be stored in encrypted form only.

3.5 Technology stack

All website development should use the following technology stack, where applicable:

- HTML Living Standard
- CSS latest stable version as defined by W3C CSS Snapshots
- Bootstrap 5 user interface framework
- Client-side JavaScript
- Server-side PHP version 7+
- MYSQL dB 5.7+.

3.6 Development environment

All software code is to be developed using a web-suitable integrated development environment (IDE) (such as WampServer or similar) that supports client-side scripting, server-side scripting and database administration.

3.7 Coding conventions

For detailed information on coding conventions and standards, refer to the Gelos Coding and Testing Guidelines (pdf).

3.8 Cyber security and software testing requirements

All software code developed must be tested sufficiently, including:

1. Functional testing
2. Non-functional testing
3. Cyber security testing including vulnerability and SQL injection
4. Compatibility testing across devices and browsers.

Testing should be documented and follow prescribed test criteria.

For more detailed information on testing code, refer to the Gelos Coding and Testing Guidelines (pdf).

3.9 File archiving and backup

All digital assets produced during work undertaken on behalf of the organisation must be backed up. Backed up files must use the file archival scheme as specified by the organisation from time to time.

File archiving is the process of collecting multiple files into a single file (often compressed) for storage, backup, transfer, or long-term preservation.

For more information on performing backups, refer to the Gelos Data Backup Policy (pdf).

4. Document information and review

4.1 Responsibilities

The following table provides a list of the positions or departments that are responsible for this policy and procedure and the scope of their responsibility.

Table 1 – Responsibilities

Position	Responsibility
CEO	Approver for this policy and procedure.

4.2 Definitions

The following definitions clarify the terms and words that are specific to this policy and procedure.

Table 2 – Definitions

Term	Definition
Scope	The extent of the area that something applies to.
Best practice guidelines	A set of criteria generally within an industry relating to best practice ways of carrying out of operations.
Organisational procedures	Clear criteria for decision making processes and the way that work in an organisation should be carried out.

4.3 Document review

This policy and procedure will be reviewed every 12 months.

Review date: 08/08/20XX

4.4 Document authority

This Website Development Policy and Procedure has been authorised by Catherine Dunn the CEO of Gelos Enterprises and is available to all staff. It has been developed in line with all relevant legislation, in consultation with committee representatives and will be revised on a regular basis.

Approval date: 08/08/20XX